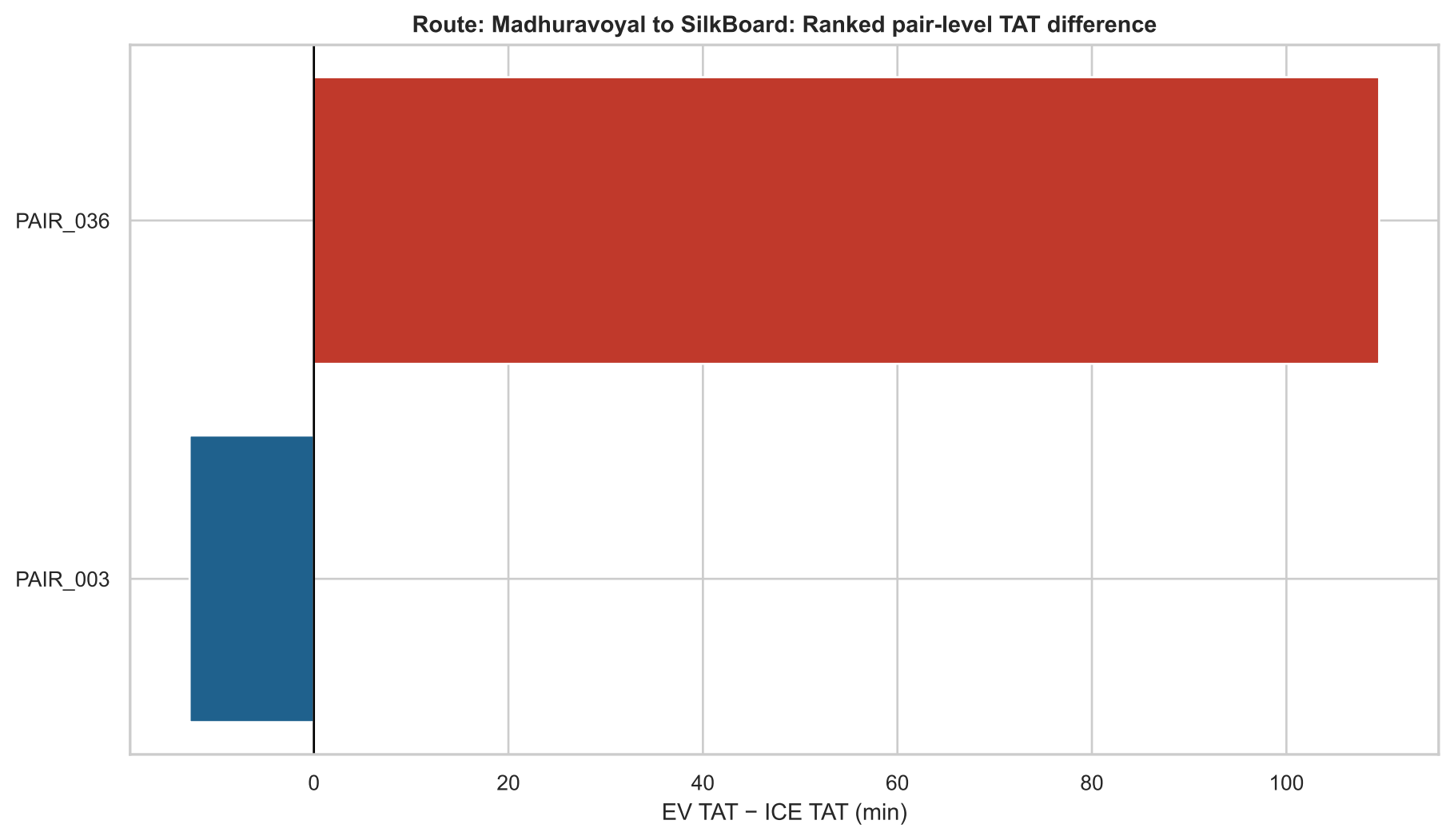
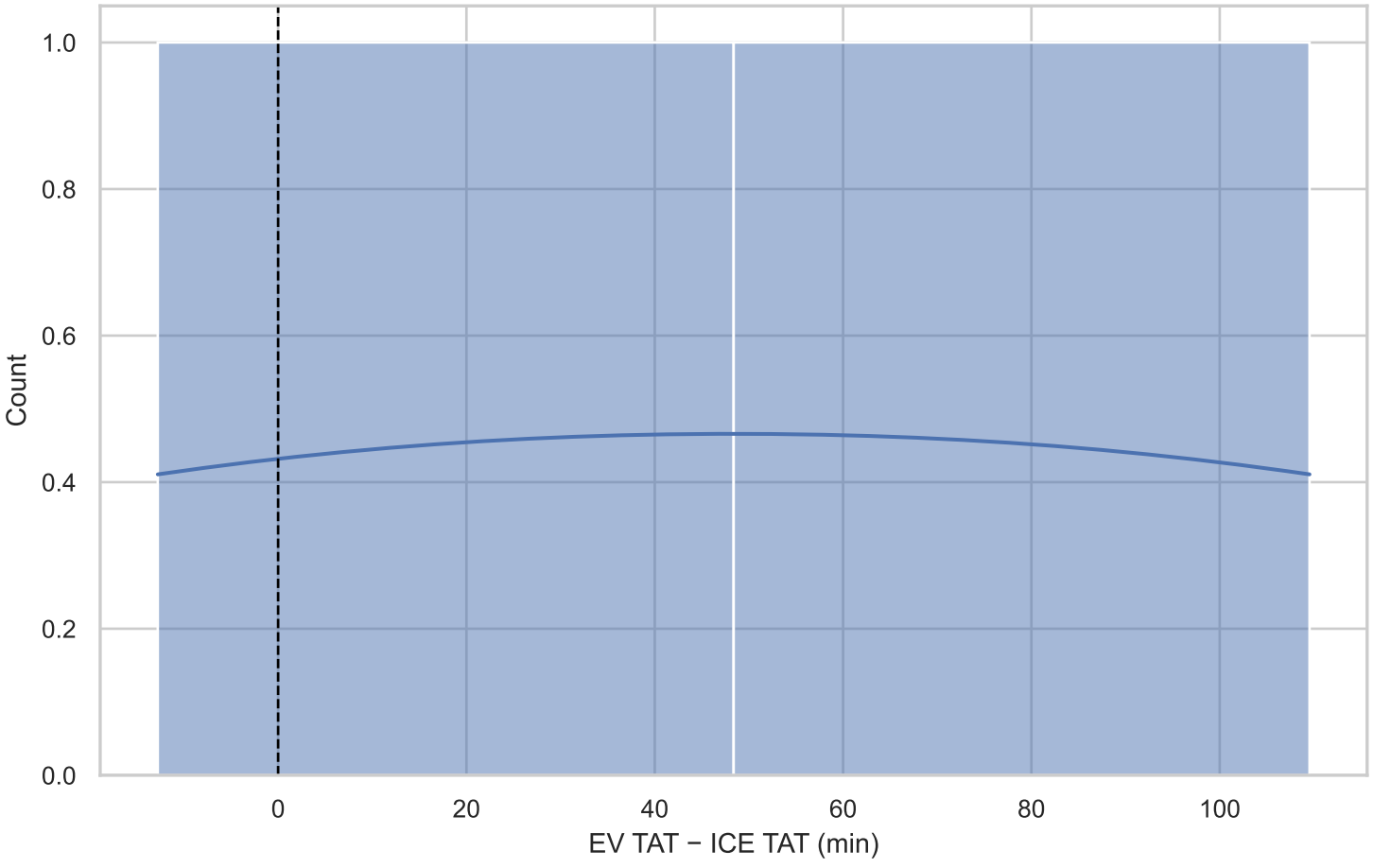
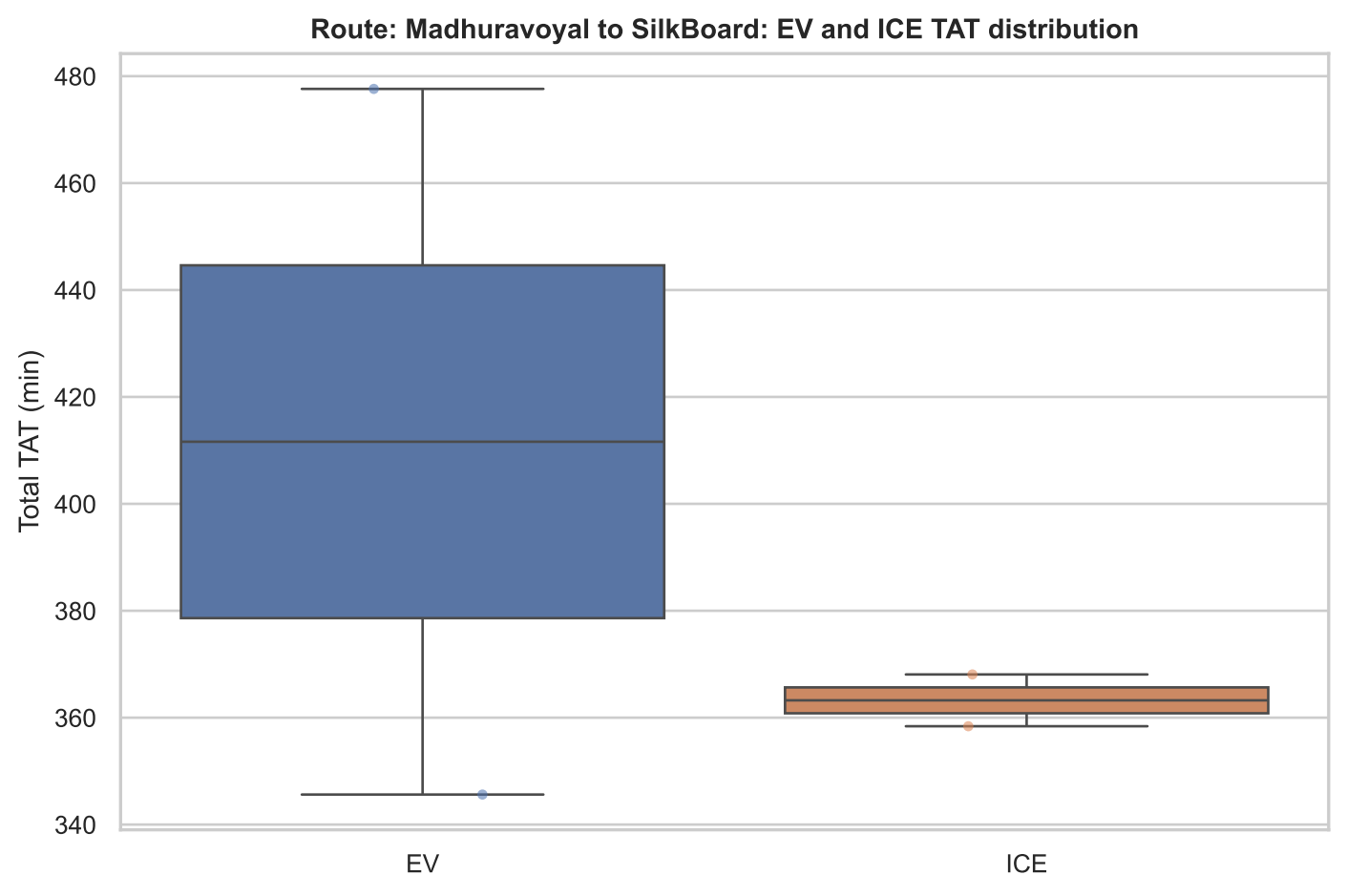
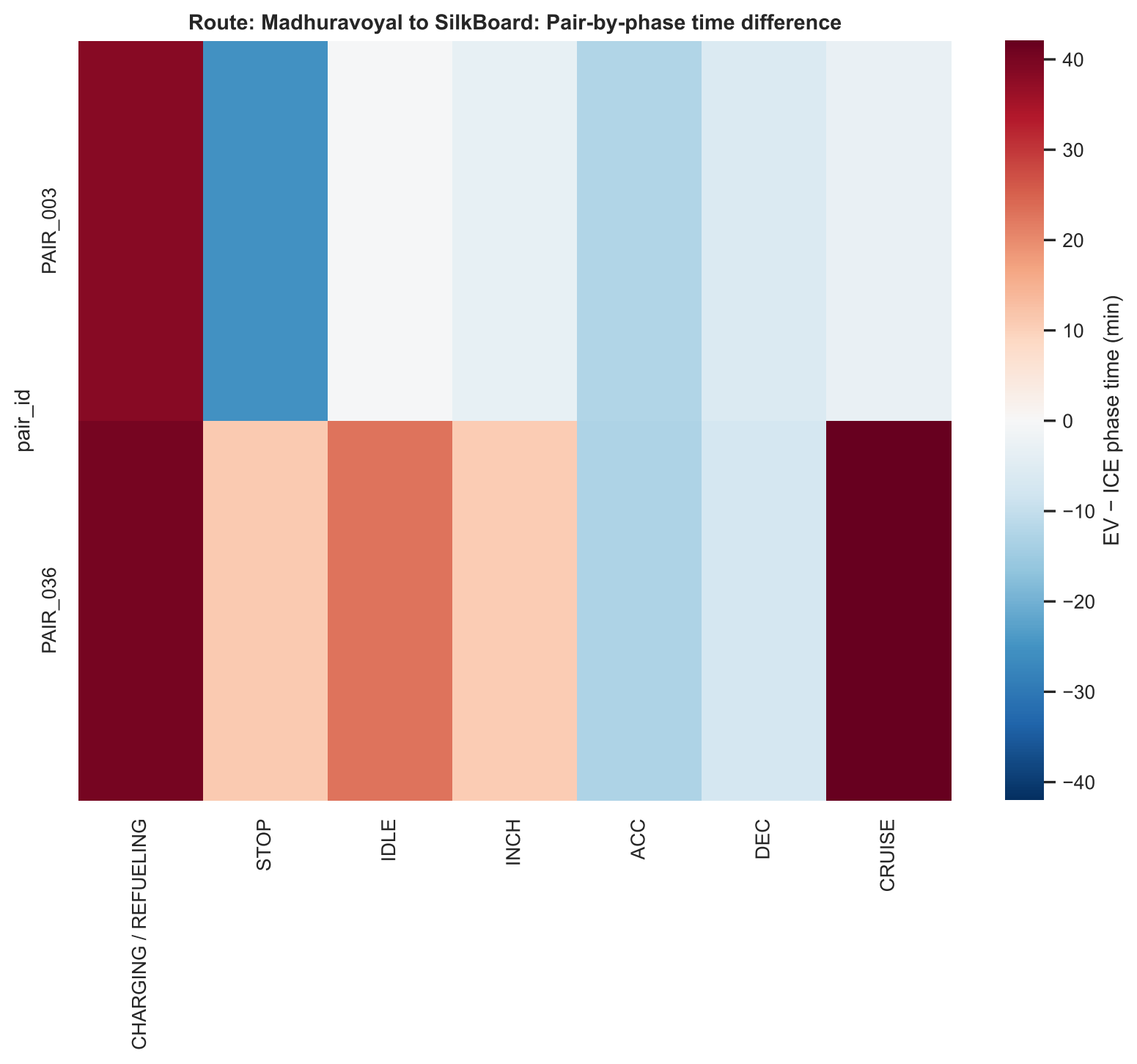
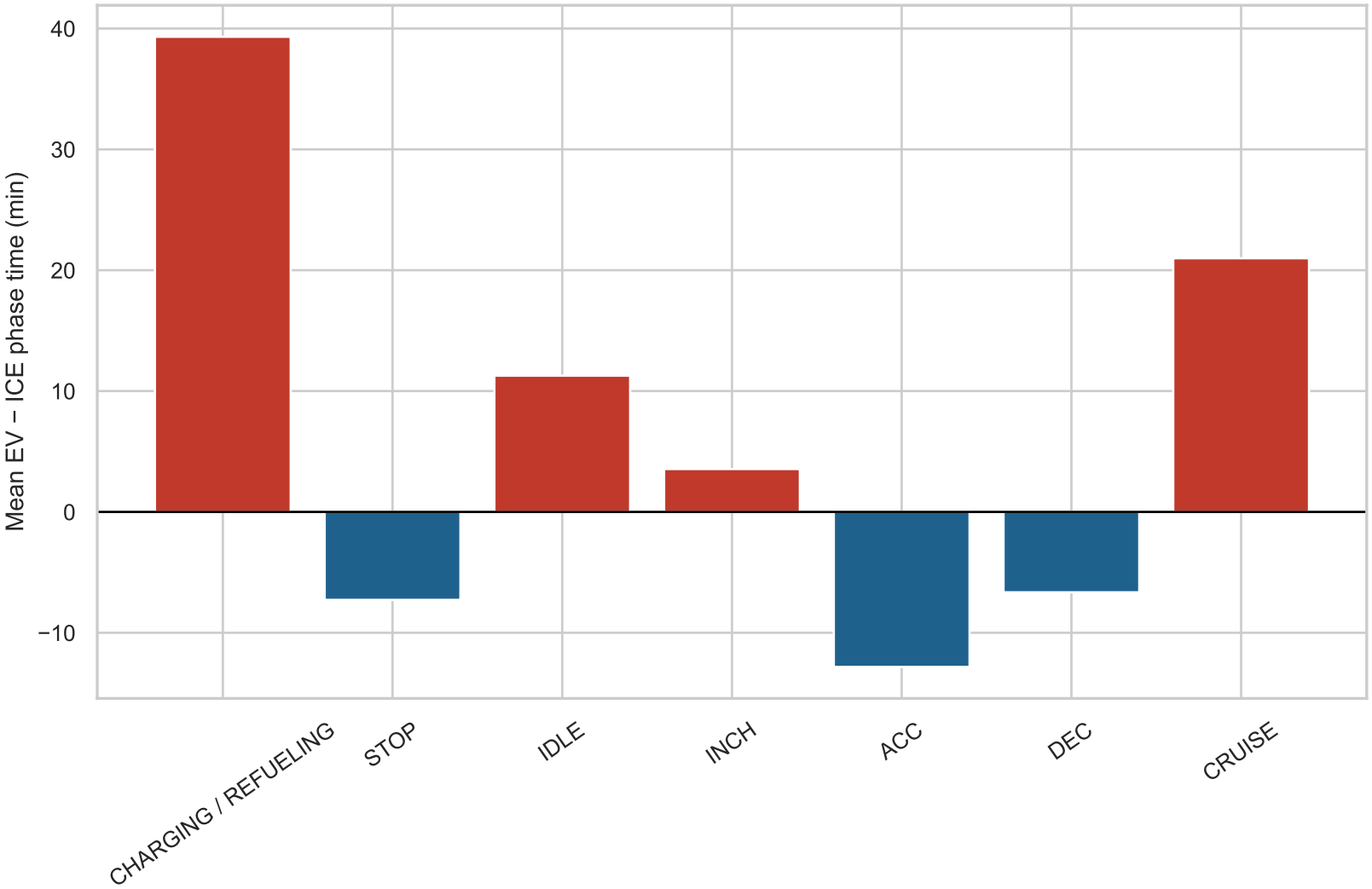
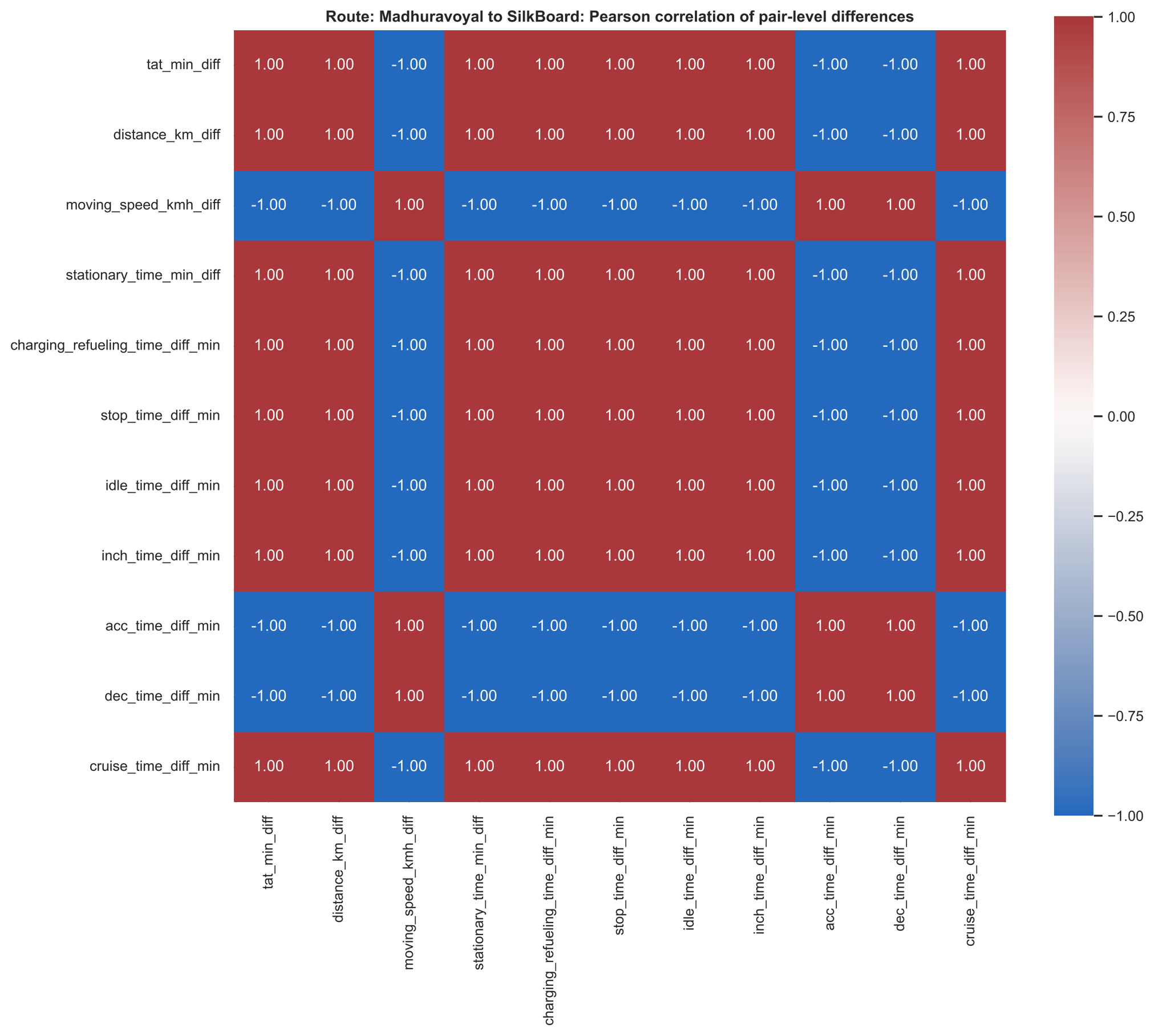

Route: Madhuravoyal to SilkBoard: Ranked pair-level TAT difference
PAIR_036
PAIR_003
0
20
40
60
80
100
EV TAT − ICE TAT (min)

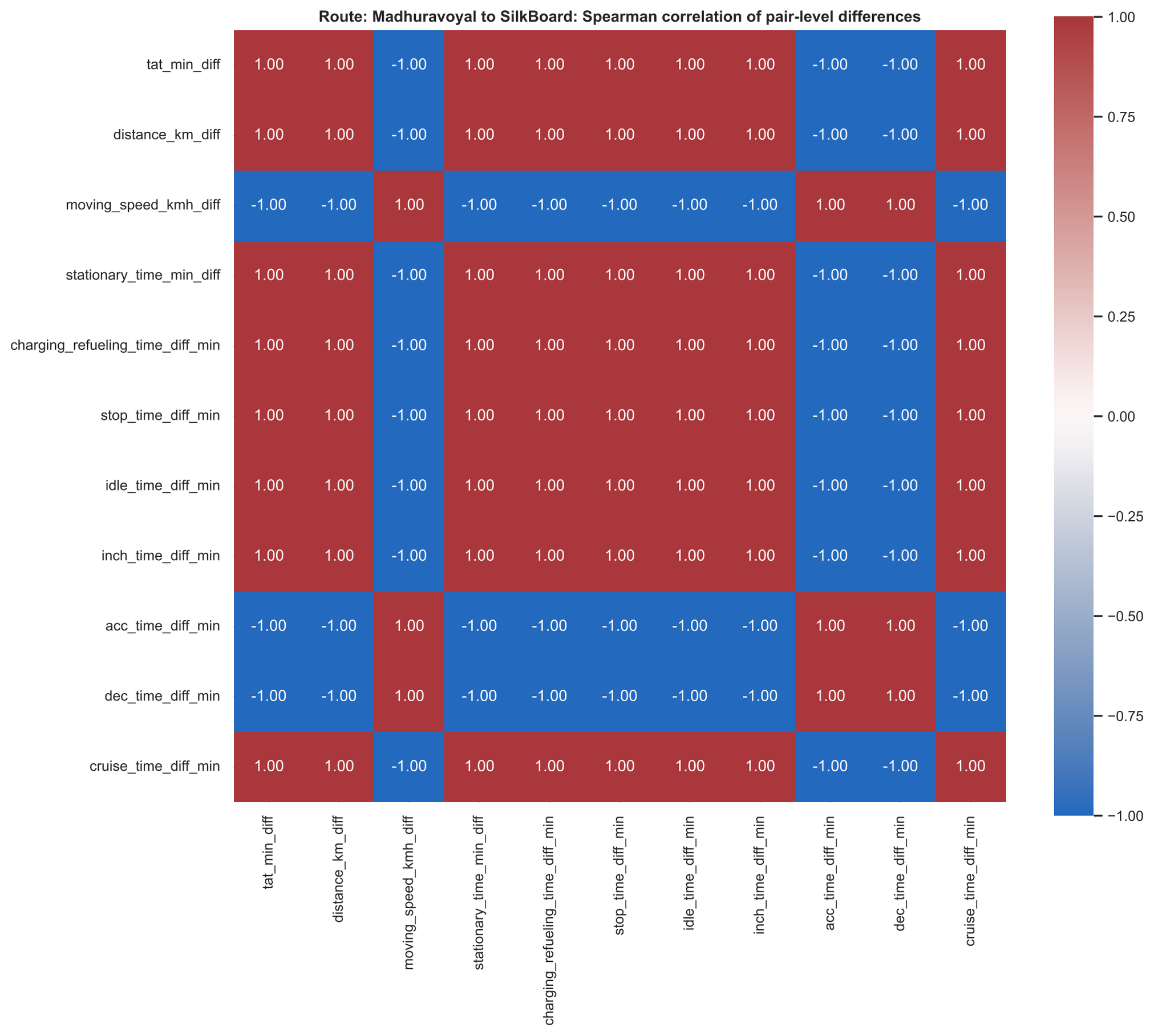

Route: Madhuravoyal to SilkBoard: Distribution of TAT differences
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
EV TAT − ICE TAT (min)

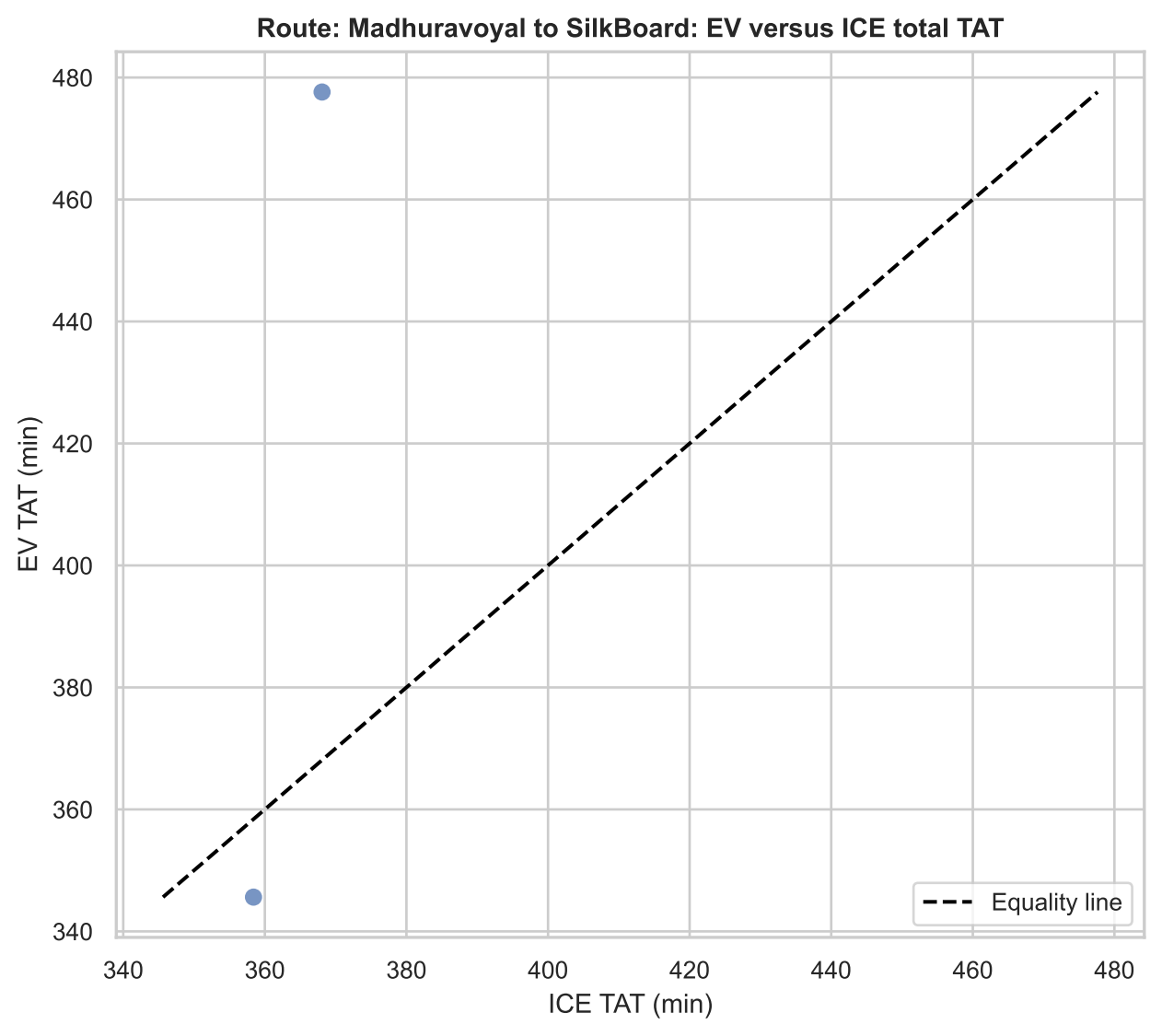

Route: Madhuravoyal to SilkBoard: EV and ICE TAT distribution
Total TAT (min)
480
460
440
420
400
380
360
340
EV
ICE

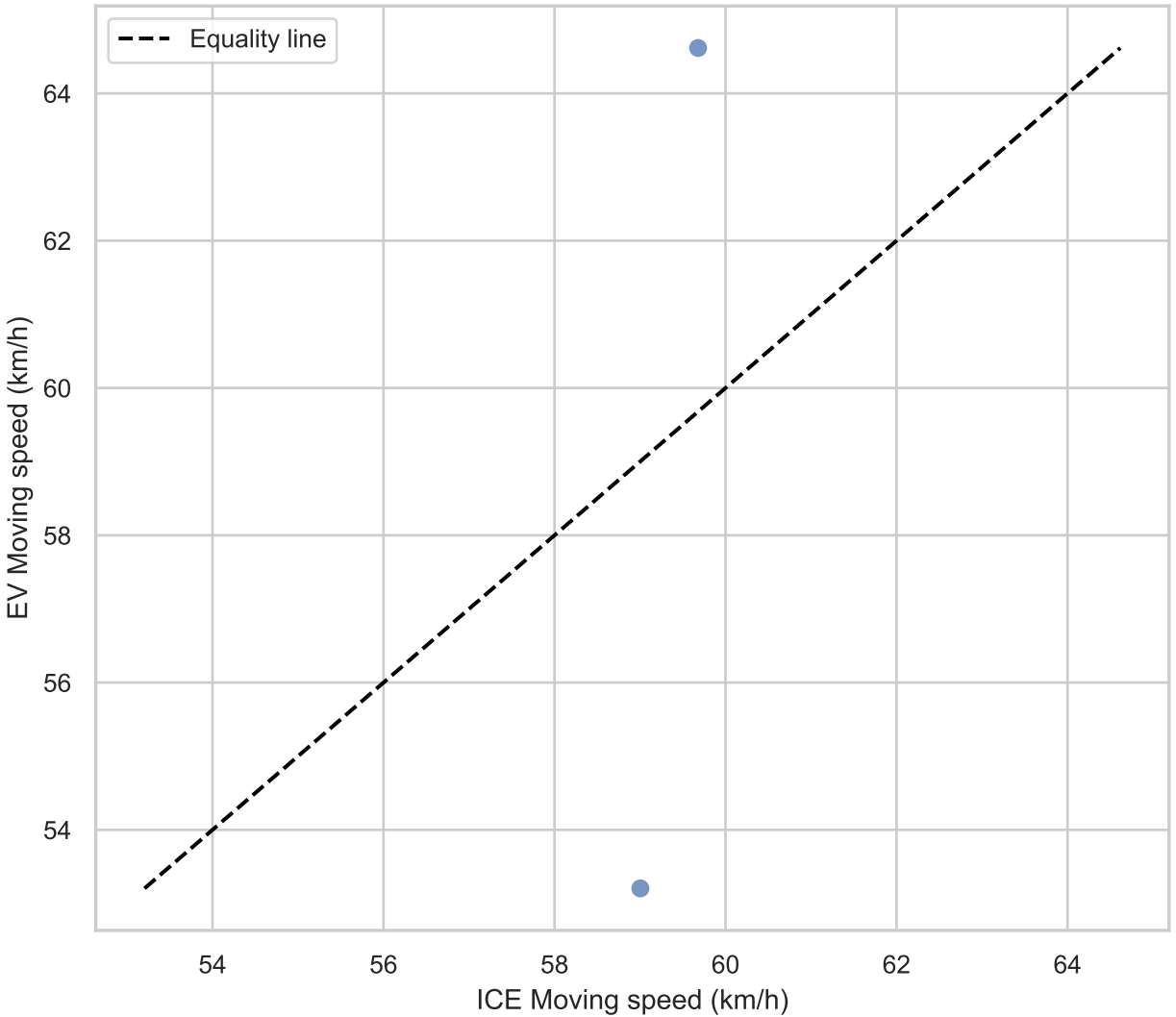

Route: Madhuravoyal to SilkBoard: Pair-by-phase time difference
pair_id
PAIR_003
PAIR_036
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
EV − ICE phase time (min)
40
30
20
10
0
−10
−20
−30
−40

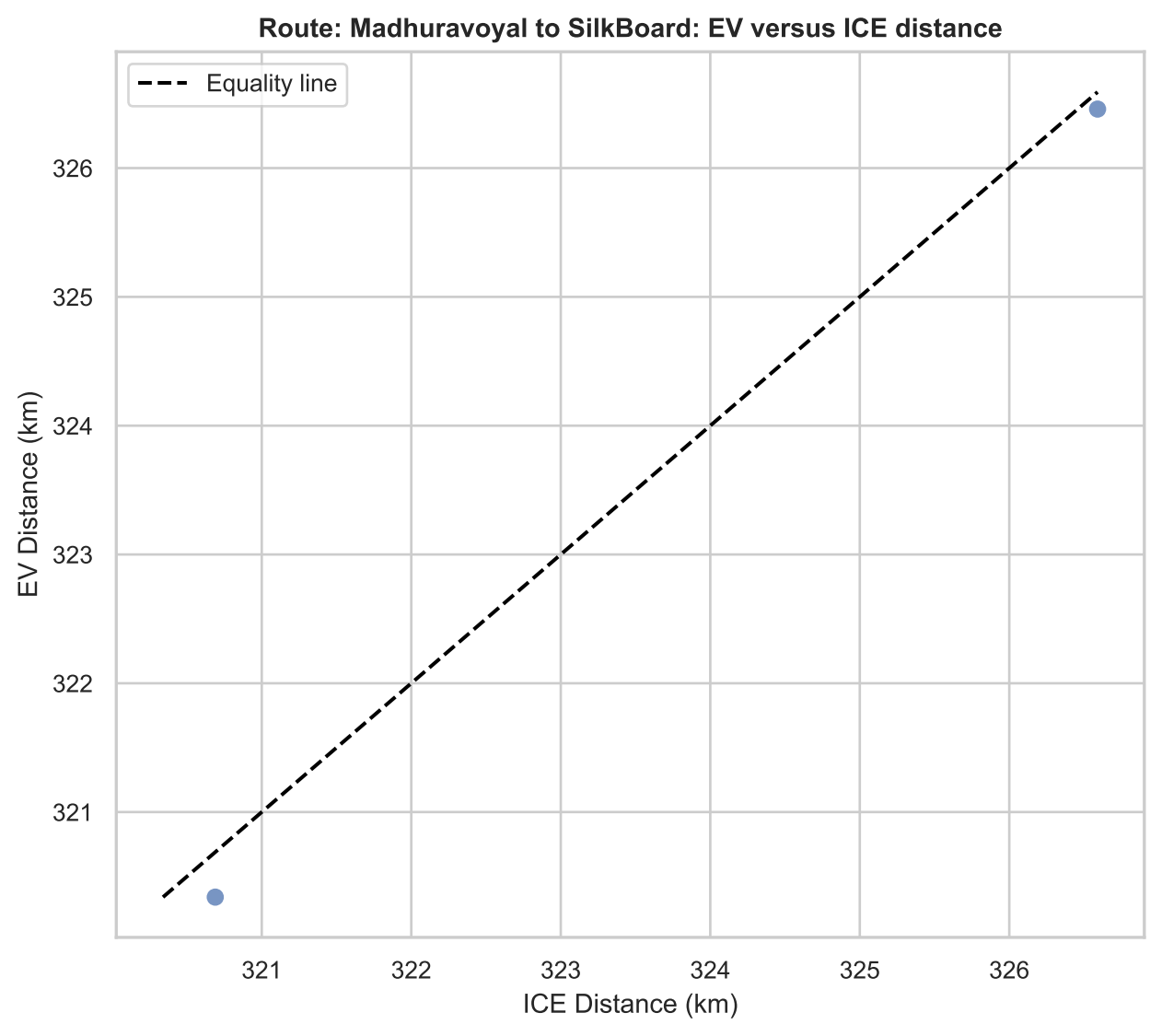

Route: Madhuravoyal to SilkBoard: Mean phase contribution to TAT difference
Mean EV – ICE phase time (min)
40
30
20
10
0
−10
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

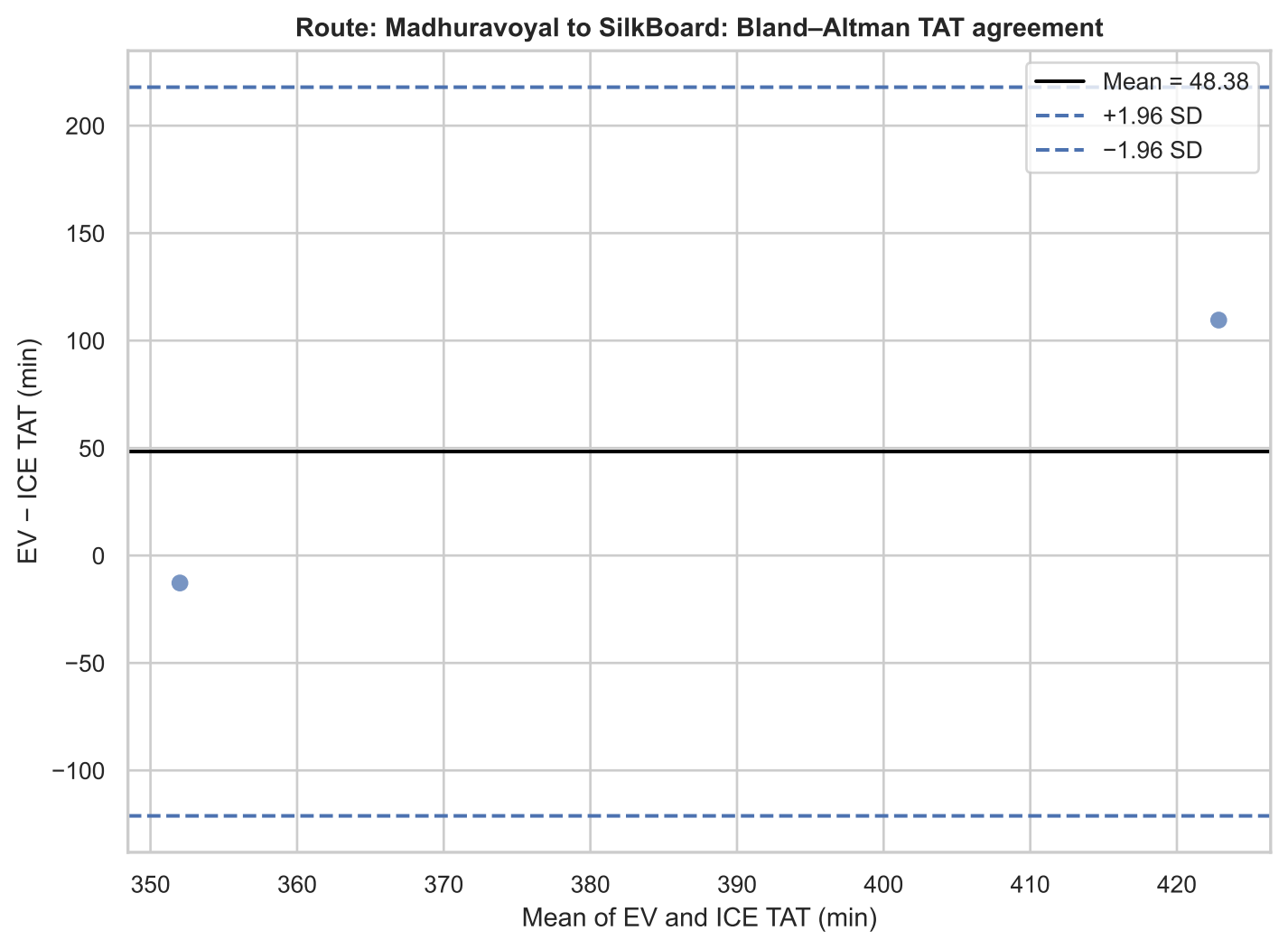

Route: Madhuravoyal to SilkBoard: Pearson correlation of pair-level differences
tat_min_diff 1.00 1.00 -1.00 1.00 1.00 1.00 1.00 1.00 -1.00 -1.00 1.00
distance_km_diff 1.00 1.00 -1.00 1.00 1.00 1.00 1.00 1.00 -1.00 -1.00 1.00
moving_speed_kmh_diff -1.00 -1.00 1.00 -1.00 -1.00 -1.00 -1.00 -1.00 1.00 1.00 -1.00
stationary_time_min_diff 1.00 1.00 -1.00 1.00 1.00 1.00 1.00 1.00 -1.00 -1.00 1.00
charging_refueling_time_diff_min 1.00 1.00 -1.00 1.00 1.00 1.00 1.00 1.00 -1.00 -1.00 1.00
stop_time_diff_min 1.00 1.00 -1.00 1.00 1.00 1.00 1.00 1.00 -1.00 -1.00 1.00
idle_time_diff_min 1.00 1.00 -1.00 1.00 1.00 1.00 1.00 1.00 -1.00 -1.00 1.00
inch_time_diff_min 1.00 1.00 -1.00 1.00 1.00 1.00 1.00 1.00 -1.00 -1.00 1.00
acc_time_diff_min -1.00 -1.00 1.00 -1.00 -1.00 -1.00 -1.00 -1.00 1.00 1.00 -1.00
dec_time_diff_min -1.00 -1.00 1.00 -1.00 -1.00 -1.00 -1.00 -1.00 1.00 1.00 -1.00
cruise_time_diff_min 1.00 1.00 -1.00 1.00 1.00 1.00 1.00 1.00 -1.00 -1.00 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

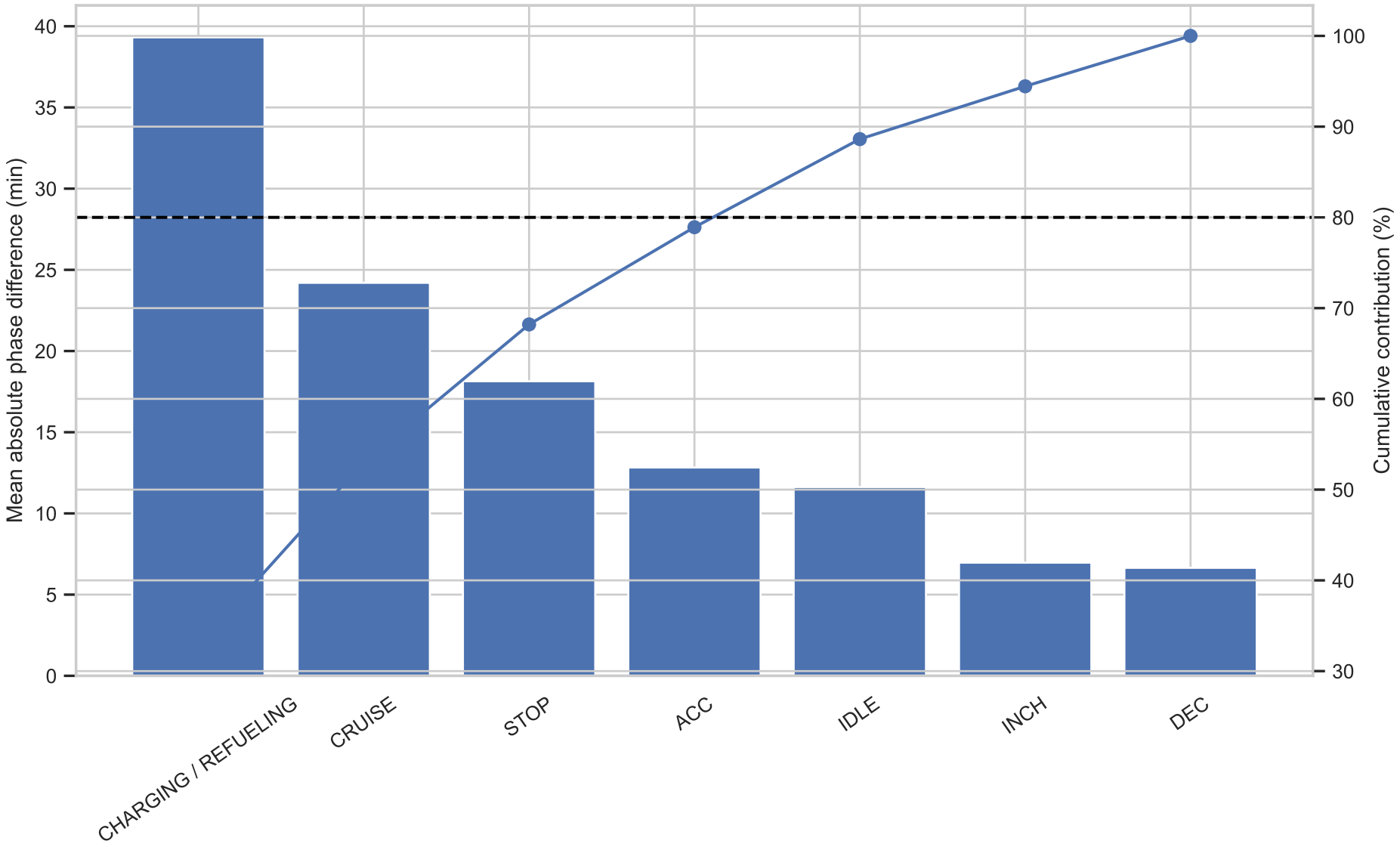

Route: Madhuravoyal to SilkBoard: Spearman correlation of pair-level differences
tat_min_diff 1.00 1.00 -1.00 1.00 1.00 1.00 1.00 1.00 -1.00 -1.00 1.00
distance_km_diff 1.00 1.00 -1.00 1.00 1.00 1.00 1.00 1.00 -1.00 -1.00 1.00
moving_speed_kmh_diff -1.00 -1.00 1.00 -1.00 -1.00 -1.00 -1.00 -1.00 1.00 1.00 -1.00
stationary_time_min_diff 1.00 1.00 -1.00 1.00 1.00 1.00 1.00 1.00 -1.00 -1.00 1.00
charging_refueling_time_diff_min 1.00 1.00 -1.00 1.00 1.00 1.00 1.00 1.00 -1.00 -1.00 1.00
stop_time_diff_min 1.00 1.00 -1.00 1.00 1.00 1.00 1.00 1.00 -1.00 -1.00 1.00
idle_time_diff_min 1.00 1.00 -1.00 1.00 1.00 1.00 1.00 1.00 -1.00 -1.00 1.00
inch_time_diff_min 1.00 1.00 -1.00 1.00 1.00 1.00 1.00 1.00 -1.00 -1.00 1.00
acc_time_diff_min -1.00 -1.00 1.00 -1.00 -1.00 -1.00 -1.00 -1.00 1.00 1.00 -1.00
dec_time_diff_min -1.00 -1.00 1.00 -1.00 -1.00 -1.00 -1.00 -1.00 1.00 1.00 -1.00
cruise_time_diff_min 1.00 1.00 -1.00 1.00 1.00 1.00 1.00 1.00 -1.00 -1.00 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

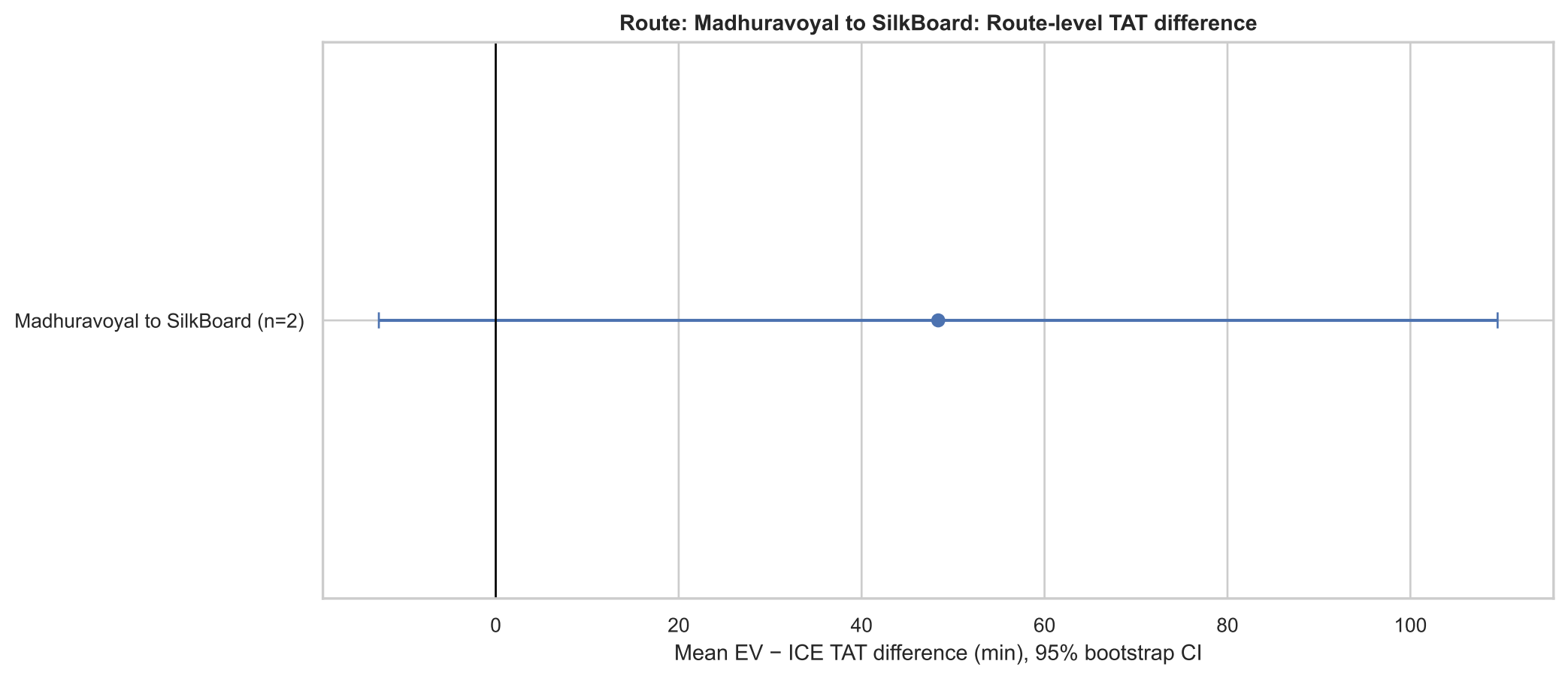

Route: Madhuravoyal to SilkBoard: EV versus ICE total TAT
EV TAT (min)
ICE TAT (min)
480
460
440
420
400
380
360
340
340
360
380
400
420
440
460
480
Equality line

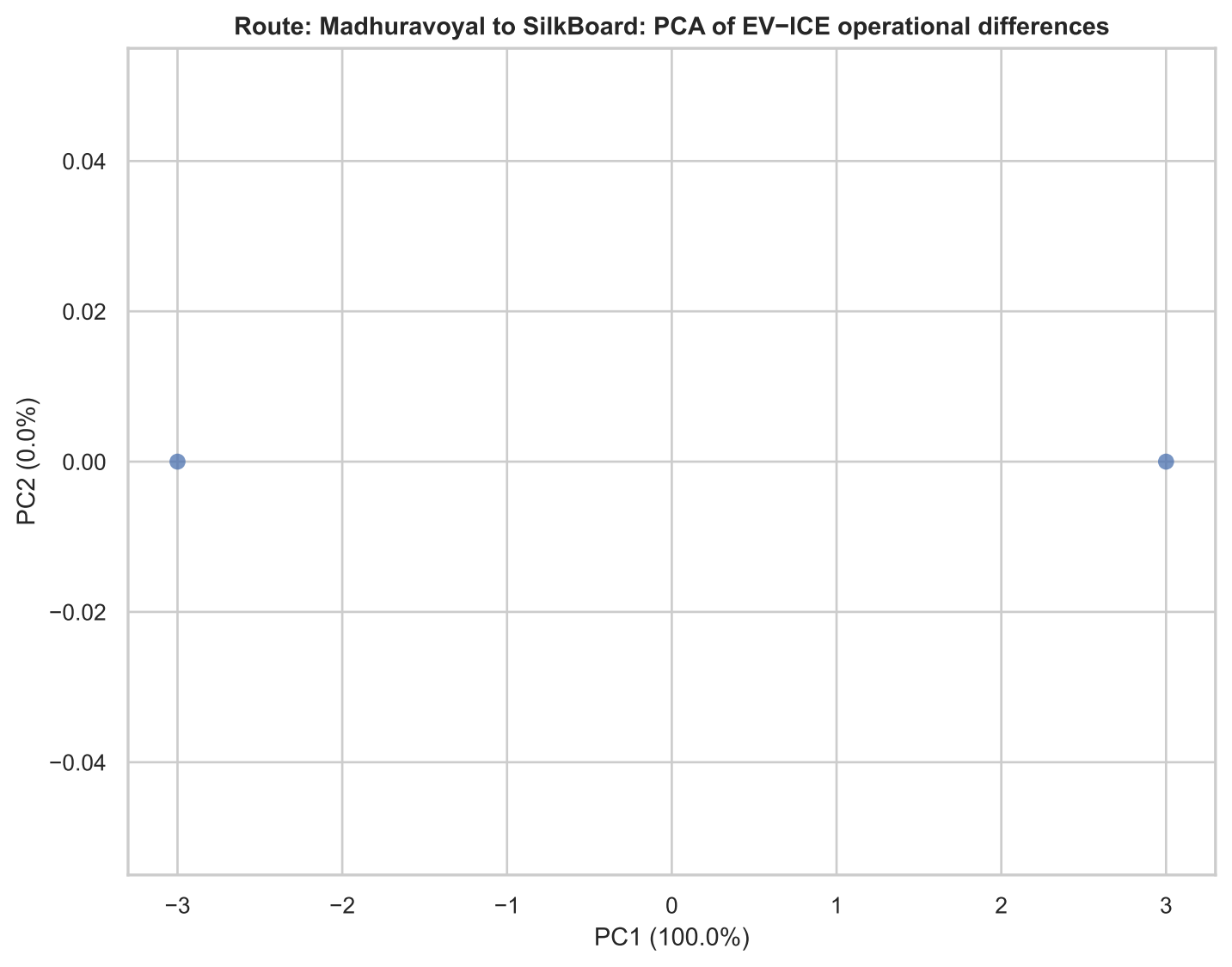

Route: Madhuravoyal to SilkBoard: EV versus ICE moving speed
Equality line
EV Moving speed (km/h)
ICE Moving speed (km/h)
54
56
58
60
62
64

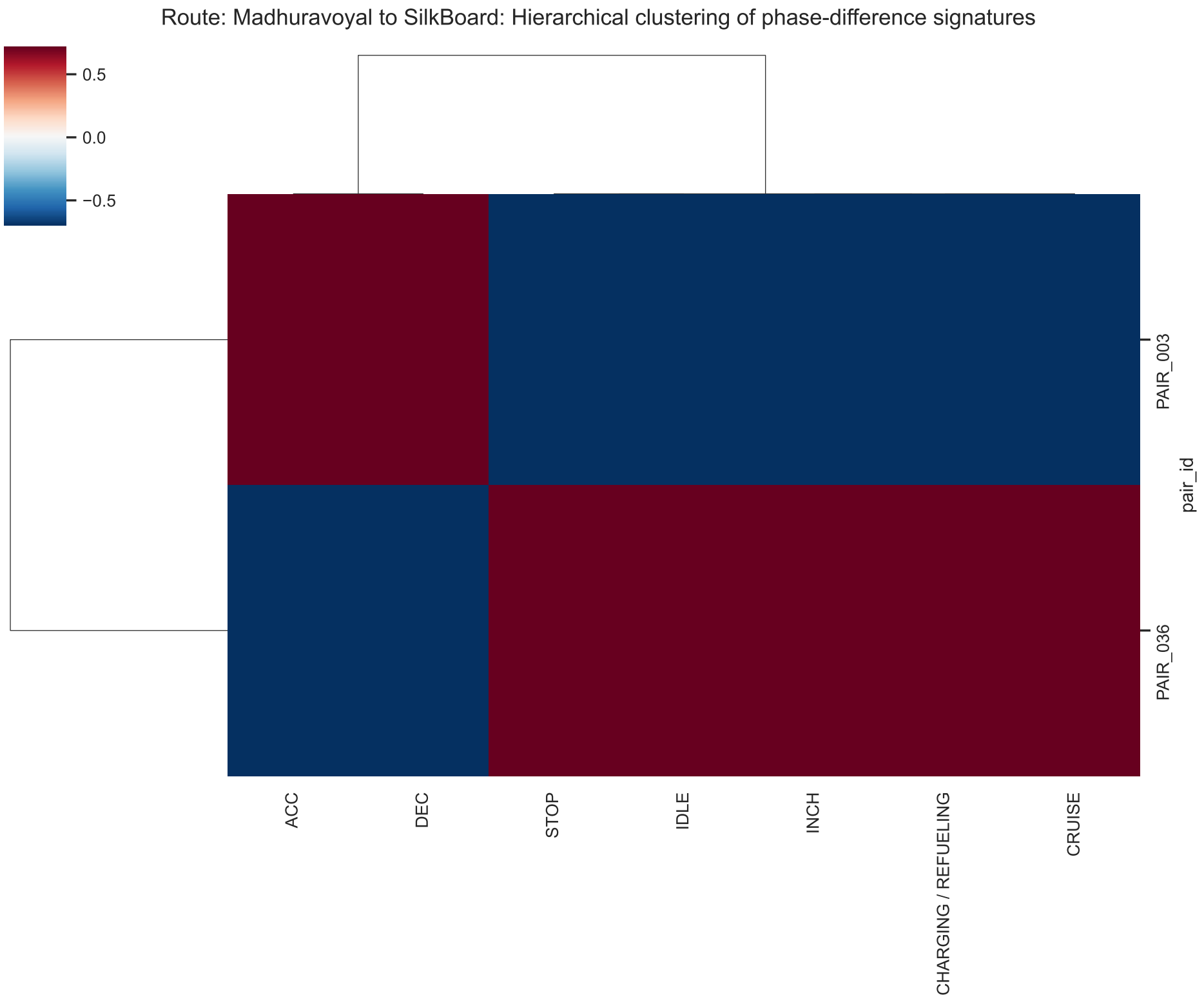

Route: Madhuravoyal to SilkBoard: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
321
322
323
324
325
326


Route: Madhuravoyal to SilkBoard: Bland–Altman TAT agreement
Mean = 48.38
+1.96 SD
−1.96 SD
200
150
100
50
0
−50
−100
EV − ICE TAT (min)
350
360
370
380
390
400
410
420
Mean of EV and ICE TAT (min)


Route: Madhuravoyal to SilkBoard: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
0
5
10
15
20
25
30
35
40
30
40
50
60
70
80
90
100
CHARGING / REFUELING
CRUISE
STOP
ACC
IDLE
INCH
DEC


Route: Madhuravoyal to SilkBoard: Route-level TAT difference
Madhuravoyal to SilkBoard (n=2)
0
20
40
60
80
100
Mean EV − ICE TAT difference (min), 95% bootstrap CI


Route: Madhuravoyal to SilkBoard: PCA of EV−ICE operational differences
0.04
0.02
0.00
−0.02
−0.04
PC2 (0.0%)
−3
−2
−1
0
1
2
3
PC1 (100.0%)


Route: Madhuravoyal to SilkBoard: Hierarchical clustering of phase-difference signatures
0.5
0.0
−0.5
PAIR_003
PAIR_036
pair_id
ACC
DEC
STOP
IDLE
INCH
CHARGING / REFUELING
CRUISE